

SEMESTER/YEAR : III SEM /II

YEARCOURSE CODE : 25MBAR001

TITLE OF THE COURSE : DATA SCIENCE FUNDAMENTALS

L: T: P: C : 3: 0: 2: 4

Overview

Decision making is essential part of a life well lived, particularly concerning business and any organization that creates customer value and so with the highest quality, lowest cost and in quick time to stay competitive and survive. In a rapidly evolving condition of market expectations, dynamic competitive, economic and political environment, decisions have to be taken evaluating various courses of possible actions and deciding on the most optimal and minimizing the risk at the same time. This calls for responsive organisations supported by responsive managers who sense reality in real time aided by quality data transformed into information, Insights and intelligence for action. There is no paucity of tools that claim to do all that one needs. It is not about the tools, it is about gauging a given business situation, understanding the problems and issues rightly so as to take the desired action. Data, Data gathering, Data organisation to draw insights to make actionable sense and then follow through to the end is the reason behind data science. Data and the science that creates business insights is the joint responsibility of business executives as well as IT teams or Data Scientists and Data Engineers. Business executives need to state requirements for data scientists to organize the plumbing required.

Course Objectives

1. To enable students to understand the role of data in decision making
2. To enable students to understand data from multiple perspective
3. To understand the data needs of various business domains and corresponding data sources
4. To derive actionable insights from data
5. To enable students to understand the importance of data for executive action to join hands with IT Teams to implement data streams from sources, create data structures to store data and be able to harness insights from such stores.

Course Outcomes

1. Exhibit the fair understanding of concepts, theories, models and principles of Data Science course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of Data Science in real-time problem-solving course.
3. Demonstrate the ability of analysing the real time problems using skill acquired by concepts, theories, models and principles of Master data management and Governance course.
4. Exhibit the ability of designing and redesigning existing and new models/solutions by using the acquired skills from concepts, theories, models and principles of Data bases, data warehouse/Data Marts, Data Mining, Business Intelligence, OLAP in real-time

problem-solving course.

5. Exhibit the ability of developing / redesigning existing and new models/solutions by using the acquired skills from concepts, theories, models and principles of Analytics and AI in real-time problem-solving course.

Approach to Learning	• Lectures • Readings • Active student participation and class room exercises • Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: • Mini projects, • Submission of assignments • Group assignments • Written Exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	INTRODUCTION • Data is the raw material for business decisions. • Data & Data Science. • Basis for Management Decisions. • Risks is embedded in all decisions, the impact and effectiveness in managing are the key. • Decisions have to be evaluated before implementation. • Data needs the sensory perception in real time for speedy and yet effective management decisions. • Data management and effective decision making have business value and they provide competitive advantage. • Role of Business Executives in defining information requirement for management and the data needed to derive information. • Role of Business Executives in the success of science of data management for their own effectiveness.	8

Unit II	All about Data ● Data sources including IoT. ● Data gathering from transaction sources and other textual or unstructured data and Big data. ● Problems with data sources and means to filter data. ● Metadata. ● Data Models, ERD and NoSQL. ● Schema. ● Conceptual, Logical and Physical Data Models. ● Organizing Data for reports, presentations, queries with speed of response. ● Data bases, Relational, NoSQL, Real and Synthetic data.	8
Unit III	Data Organization ● Data Structures. ● Data Bases. ● Data Warehouses and Data Marts and ETL. ● organization of data for Data Mining, BI and OLAP. Data Presentation	8
Unit IV	Data to Insights – Transformational processes ● ETL in Depth. ● Business statistics. ● Descriptive, Predictive and Prescriptive Analytics. ● Analytics models, graphical, Algebraic and Spreadsheet models. Linear, Integer and non-linear optimization.	8
Unit V	Data Science Driven Applications ● AI & ML ● Robotics ● RPA ● Digitally Enabled Future. Imagination is the only limit. Plenty of opportunities. ● Data Science Driven application – State of the art with real world success cases. ● Data is the key to business consciousness without it any business is dead. ● There is no good business management without corresponding data management.	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO1	3	-	-	2	2
CO2	3	-	-	-	2
CO3	3	-	-	-	2
CO4	3	-	-	2	2
CO5	3	-	-	1	2

Action Based Component

- Case Studies, Presentations, Simulations, Field projects, Assignments

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X	X			
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Data Science, Richard Hurley, MIT Press (Essential Knowledge Series)
2. Data Science for Business, Thomas Provost & Tom Fawcett

Reference books

1. Master Data Management & Data Governance, Alex Berson & Larry Dubov – McGraw Hill.
2. Introduction to Data Mining with Case studies, GK Gupta - PHI.
3. Data Warehousing in the Real World, Sam Anahory & Dennis Murray – Pearson.
4. Business Analytics, James Evans – Pearson.

5. Wayne Winston, Microsoft Excel Data Analysis and Business Modeling, 5th Edition, Microsoft Press.
6. Davenport Thomas H., Competing on Analytics: Updated, with a New Introduction, Harvard Business School Press.

Readings & Case Analysis

1. Harvard Business Review
2. MIT Sloan Management Review

SEMESTER/YEAR : III SEM /II

YEARCOURSE CODE : 25MBAR002

**TITLE OF THE COURSE : AI BASICS: MACHINE LEARNING, DEEP LEARNING,
NLP, RI**

L: T: P: C : 3: 0: 2: 4

Overview

Deep Learning is creating a lot of buzz in the Artificial Intelligence landscape. Students will understand different types of deep learning models such as SLP, MLP, RNN, CNN, RBM etc. They will also be able to know which model is suitable for which type of tasks. They will be able to use appropriate libraries in Python to implement these neural networks.

Course Objectives

Learn the fundamentals of Deep Learning

1. Understand the business context to Deep Learning
2. Appreciate different types of Deep Learning algorithms and their application
3. Implement Supervised Machine Learning algorithms using R in a business application
4. Implement Unsupervised Machine Learning algorithms using R in a business application
5. Evaluate Model Performance

Course Outcomes

By the conclusion of this course, the student should be able to

1. Exhibit the fair understanding of concepts, theories, models and principles of deep learning course
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of deep learning in real-time problem-solving course
3. Demonstrate the ability of analysing the real time problems using skill acquired by concepts, theories, models and principles of artificial neural network course
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of convolutional neural network in real-time problem-solving course
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of recurring neural network and other deep learning models in real-time problem-solving course

Approach to Learning	• Lectures • Readings • Active student participation and classroom exercises • Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: • Mini projects, • Submission of assignments • Group assignments • Written Exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	Introduction to Deep Learning • Deep Learning Models • Single Layer Perceptron Model (SLP) • Multilayer Perceptron Model (MLP) • Convolutional Neural Networks (CNNs) • Recurrent Neural Networks (RNNs) • Restricted Boltzmann Machines (RBMs) • Deep Belief Networks (DBNs) • Experimental Design • Feature Selection • Applied Machine Learning and Deep Learning	8
Unit II	Single and Multilayer Perceptron Models • SLP Model • Training the Perceptron Model • Widrow-Hoff Algorithm • Limitations of SLP • MLP Model • Converging upon a global optimum • Back propagation algorithm for MLP • How many hidden layers	8
Unit III	Convolutional Neural Networks • Structure and property of CNNs • Components of CNN Architectures • Convolutional Layer • Pooling Layer • Rectified Linear Units (ReLU) Layer • Fully Connected (FC) Layer • Loss Layer	8

Unit IV	Recurrent Neural Networks - Fully Recurrent Networks - Training RNNs with Back-Propagation Through Time (BPPT) - Elman Neural Networks - Neural History Compressor - Long Short-Term Memory (LSTM) - Traditional LSTM - Training LSTMs - Structural Damping Within RNNs - Tuning Parameter Update Algorithm - Practical Example of RNN: Pattern Detection	8
Unit V	Autoencoders, Restricted Boltzmann Machines, and Deep Belief Networks - Autoencoders - Linear Autoencoders vs. Principal Components Analysis (PCA) - Restricted Boltzmann Machines - Contrastive Divergence (CD) Learning - Momentum Within RBMs - Weight Decay - Sparsity - Deep Belief Networks (DBNs) - Fast Learning Algorithm (Hinton and Osindero 2006)	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO1	3	-	-	-	2
CO2	3	-	-	-	2
CO3	3	-	-	-	2
CO4	3	-	-	-	2
CO5	3	-	-	-	2

Action Based Component

- Case Studies, Presentations, Simulations, Field projects, Assignments

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X	X			
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Introduction to Deep Learning using R (Apress), Taweh Beysolow

Reference books

1. <https://cognitiveclass.ai/courses/introduction-deep-learning>
2. <https://www.deeplearningbook.org/>

Readings & Case Analysis

1. Harvard Business Review
2. MIT Sloan Management Review

SEMESTER/YEAR : III SEM /II

YEARCOURSE CODE : 25MBAR003

TITLE OF THE COURSE : PYTHON PROGRAMMING AND PYTORCH

L: T: P: C : 3: 0: 2: 4

Overview

In this course, students will learn basics of machine learning using Python programming. They will get a perspective on

- Where Machine Learning fits into Analytics landscape
- Types of Machine Learning algorithms
- Data Science process from data collection till model evaluation

Course will be hands on Jupyter Notebook (Anaconda) and mini projects will be implemented for each algorithm type.

Course Objectives

1. Understand the business context to Machine Learning
2. Appreciate different types of Machine Learning algorithms and their application
3. Learn Python Data Structures
4. Learn Exploratory Data Analysis and its significance
5. Implement Supervised & Unsupervised Machine Learning algorithms using Python in a business application

Course Outcomes

By the conclusion of this course, the student should be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of data architecture course
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of supervised and unsupervised ML algorithms in real-time problem-solving course
3. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of machine learning in real-time problem-solving course
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of machine learning in real-time problem-solving course
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of machine learning in real-time problem-solving course

Approach to Learning	• Lectures • Readings • Active student participation and class room exercises • Case Analysis collaboratively with students' involvement
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Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam
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Syllabus

<u>Units</u>	<u>Syllabus Details</u>	<u>Teaching Hours</u>
Unit I	Introduction to ML ● Why Machine Learning? ● Why Python? ● Scikit-learn ● Jupyter Notebook ● Numpy ● Scipy ● Matplotlib ● Pandas ● Mglern ● First Application-classifying iris species	8
Unit II	Supervised Learning ● Classification and Regression ● Generalization, Overfitting and Underfitting ● K-Nearest Neighbours ● Linear Models ● Naive Bayes Classifier ● Decision Tree ● Ensemble of Decision Trees ● Support Vector Machines ● Neural Networks	8
Unit III	Unsupervised Learning and Preprocessing ● Types of Unsupervised Learning ● Challenges in Unsupervised Learning ● Preprocessing and scaling ● Dimensionality reduction, Feature extraction and manifold learning: PCA, NMF, t-SNE ● Clustering: k-Means, Agglomerative, DBSCAN	8
Unit IV	Representing Data and Engineering Features ● Categorical Variables: One hot encoding (Dummy Variables) ● Binning, Discretization, Linear Models and Trees ● Automatic Feature Selection: Model based feature selection, Iterative feature selection	8

Unit V	Model Evaluation and Improvement • Cross-Validation in scikit-learn • Benefits of Cross-Validation • Stratified k-fold Cross-Validation • Grid Search • Evaluation Metrics and Scoring	8
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CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO1	3	-	-	2	2
CO2	3	-	-	-	2
CO3	3	-	-	-	2
CO4	3	-	-	-	2
CO5	3	-	-	-	2

Action Based Component

- Case Studies, Presentations, Simulations, Field projects, Assignments

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X	X			
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Introduction to Machine Learning with Python (O'Reilly), Andreas C Muller & Sarah Guido

Reference books

1. <https://cognitiveclass.ai/courses/machine-learning-with-python>

Readings & Case Analysis

1. Harvard Business Review
2. MIT Sloan Management Review

SEMESTER/YEAR : III SEM /II

YEARCOURSE CODE : 25MBAR004

TITLE OF THE COURSE : MANAGING AI PROJECTS AND TRENDS IN AI

L: T: P: C : 3: 0: 2: 4

Overview

There is hardly any industry operative in the world today that is oblivious to the high revenue and value offering potential that artificial intelligence comes packed with. A declarative fact that is evident from the promising AI technology trends for 2020 and later. This quick adoption, while on one hand has come with a lot of benefits for both businesses and end-users, it is on the other hand is on a very native stage. Meaning, businesses are yetto find concrete use cases and return effectiveness. This nascency and benefit combination has given birth to a number of queries around how to manage your AI projects.

Seeing how complexity lies at the very centre of the AI project management solutions, it is important to understand the intricacies of managing AI projects. In this course, students will learn about AI project management and the steps we follow to successfully build a Proof of Value (POV) to AI solution & services.

Course Objectives

1. Study how AI projects are different from traditional projects
2. Understand the structure and basic elements of an AI project
3. Understand the success factors of an AI project
4. Understand the reasons for failure of AI projects

Course Outcomes

By the conclusion of this course, the student should be able to

1. Exhibit the fair understanding of concepts, theories, models and principles of AI transformation fundamentals and trends course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of AI-driven project strategy in real-time problem-solving course.
3. Demonstrate the ability of analysing the real time problems using skill acquired by concepts, theories, models and principles of organizational, societal, and geopolitical impacts of AI Project Management course.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of ethical AI frameworks in real-time problem-solving course.
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of global AI governance and global trends in real-time problem-solving course.

Approach to Learning

- Lectures
- Readings
- Active student participation and class room exercises
- Case Analysis collaboratively with students' involvement

Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam
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Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	How is an AI Project Different from Traditional Projects? AI project management calls for a different approach when parallels are drawn between them and traditional mobile app project management. Meaning, the differences between AI projects and traditional IT projects are manifold.	8
Unit II	Understanding the Pillars of AI Project Success Understand how the secret to an AI project success lies in two pillars – people and data. Only in the presence of the two pillars, AI is able to improve customer experience to its entirety.	8
Unit III	Why AI projects fail A list of what challenges companies face when implementing AI will be very extensive. But at the core of it all of why Proof of Values fail, lies two prime causes: misaligned expectations and insufficient data management capabilities. These are causes that hold back businesses from making money in AI. This unit provides an in depth understanding of these factors.	8
Unit IV	Stepwise approach to AI projects This course unit covers a stepwise approach to defining and implementing AI projects in an Enterprise. Similar to other software development projects, understanding customer needs is key as is the identification of the right approach and people. The students will gain an understanding of how a manager of an AI project can follow a systematic approach to create a successful AI project.	8
Unit V	Recent trends in AI Spending on research, development, and deployment continues to rise and debate over the wider social implication's rages on. Meanwhile, the incentives only get bigger for those looking to roll out AI-driven innovation into new areas of industry, fields of science, and our day-to-day lives. This course section will help students the recent advances and trends in Artificial Intelligence.	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO1	3	-	-	-	2
CO2	3	-	2	-	2
CO3	3	-	2	-	2
CO4	3	-	2	-	2
CO5	3	-	-	-	2

Action Based Component

- Case Studies, Presentations, Simulations, Field projects, Assignments

Course Assessment

#	Description of Assessment method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
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2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X	X			
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Artificial Intelligence – A Modern Approach: Stuart Russell and Peter Norvig, Pearson Publications, Third Edition, 2020.

Reference books

1. ARTIFICIAL INTELLIGENCE: Kevin Knight, Elaine Rich, B. Nair, Pearson Publications, Third Edition, July 2017.

Readings & Case Analysis

1. Harvard Business Review
2. MIT Sloan Management Review

SEMESTER/YEAR : IV SEM /II YEAR
COURSE CODE : 25MBAR005
TITLE OF THE COURSE : INDUSTRY APPLICATIONS OF AI: BFSI, HEALTH CARE, EDUCATION, MANUFACTURING, SCM, RETAIL, PHARMA
L: T: P: C : 3: 0: 2: 4

Overview

Artificial intelligence (AI) is all around us. It's how Google answers our searches, Amazon recommends products, and Pandora plays another song. Artificial Intelligence is changing the way decisions are being made in the world today. This course is intended to enable students to understand a critical aspect of AI, i.e., industry applications and scenarios and use cases. The course covers this very critical aspect of AI, i.e., use cases of AI in a variety of industries such as Sales and Marketing, Retail Manufacturing, Healthcare, Education to name a few.

Course Objectives

1. Study how specific techniques in AI have been applied to a variety of industry scenarios and how choices have been made
2. Understand how given a new scenario one can define the role of AI to solve a problem
3. Capture the scenario into a set of use cases that become requirements for an AI application

Course Outcomes

By the conclusion of this course, the student should be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of AI applications in sales and marketing course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of AI applications in e-commerce in real-time problem-solving course.
3. Demonstrate the ability of analysing the real time problems using skill acquired by concepts, theories, models and principles of AI applications in healthcare course.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of AI applications in Manufacturing real-time problem-solving course.
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of AI applications in scenario analysis in real-time problem-solving course.

Approach to Learning	• Lectures • Readings • Active student participation and classroom exercises • Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: • Mini projects, • Submission of assignments • Group assignments • Written Exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	AI in Sales and Marketing Understand how AI can help build more effective marketing strategies, improve the customer journey, and change the way businesses attract, nurture, and convert prospects. React, Act, Convert, Engage is how AI and machine learning can be incorporated into every step in the customer 's lifecycle.	8
Unit II	AI in Manufacturing Understand how AI algorithms can also be used to optimize manufacturing supply chains, helping companies anticipate market changes. This information is invaluable to manufacturers as it allows them to optimize staffing, inventory control, energy consumption and the supply of raw materials.	8
Unit III	e-Commerce and Artificial Intelligence Many e-commerce businesses are already using AI for better understanding whereas the rest have already noticed the importance of using AI in e-Commerce and in the process to adopt it. To generate more leads and to provide excellent user engagement, the e-commerce business is joining hands to artificial intelligence. The companies are using voice and chat/voice bots, personalized recommendations, forecasting, filtering fake reviews, etc.	8
Unit IV	Healthcare systems using Artificial Intelligence AI offers a number of advantages over traditional analytics and clinical decision-making techniques. Learning algorithms can become more precise and accurate as they interact with training data, allowing humans to gain unprecedented insights into diagnostics, care processes, treatment variability, and patient outcomes.	8

Unit V	Scenario analysis and AI technique selection Understand how to understand a scenario and break it down into sub problems. Understand how to compare similar scenarios which have been solved using existing AI techniques and make decisions about selecting a technique and tool. Define the scenario as a set of use cases.	8
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CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	3	2	-	2	2
CO 2	3	2	-	2	2
CO 3	3	2	-	2	2
CO 4	3	2	-	2	2
CO 5	3	-	-	1	2

Action Based Component

- Case Studies, Presentations, Simulations, Field projects, Assignments

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
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2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X	X			

6	CBT	10	X	X	X			
7	Semester E nd Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Artificial Intelligence - A Modern Approach: Stuart Russell and Peter Norvig, Pearson Publications, Third Edition, 2020.

Reference books

1. ARTIFICIAL INTELLIGENCE: Kevin Knight, Elaine Rich,
B. Nair, Pearson Publications, Third Edition, July 2017.

Readings & Case Analysis

1. Harvard Business Review
MIT Sloan Management Review